

ARCHITECTING THE HDB RESALE BOT V3.0

A SERVERLESS, GCP-NATIVE BLUEPRINT FROM
DATA INGESTION TO NATURAL DIALOGUE



SCALE 1:100 (NTS)





ARCHITECTURAL DIAGNOSTIC MATRIX: LEGACY vs. GCP NATIVE vs. STRATEGIC BENEFIT

PROJECT: SINGAPORE MODERN ESTATE - DIGITAL INFRASTRUCTURE UPGRADE
REGION: SINGAPORE
MODERN CORE: DIGITAL INFRASTRUCTURE UPGRADE

DESIGN NO: A-101, REV C

DATE: 2023-10-27

BY: [Signature]

CHECKED BY: [Signature]

APPROVED BY: [Signature]

REVISIONS:

NO. DESCRIPTION

1. Initial Design

2. Revision 1

3. Revision 2

4. Revision 3

5. Revision 4

6. Revision 5

7. Revision 6

8. Revision 7

9. Revision 8

10. Revision 9

11. Revision 10

12. Revision 11

13. Revision 12

14. Revision 13

15. Revision 14

16. Revision 15

17. Revision 16

18. Revision 17

19. Revision 18

20. Revision 19

21. Revision 20

22. Revision 21

23. Revision 22

24. Revision 23

25. Revision 24

26. Revision 25

27. Revision 26

28. Revision 27

29. Revision 28

30. Revision 29

31. Revision 30

LEGACY (v2.0)

GCP NATIVE (v3.0)

STRATEGIC BENEFIT

COMPUTE

SECTION 1: COMPUTE LAYOUT | SCALE: 1:750

VPS / Docker Compose

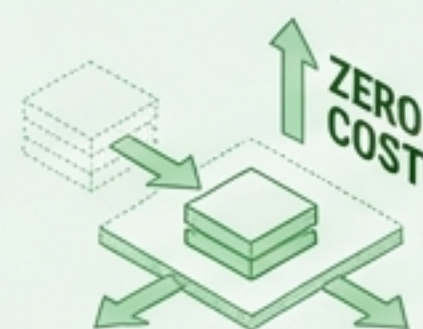


Google Cloud Run



Serverless-first.

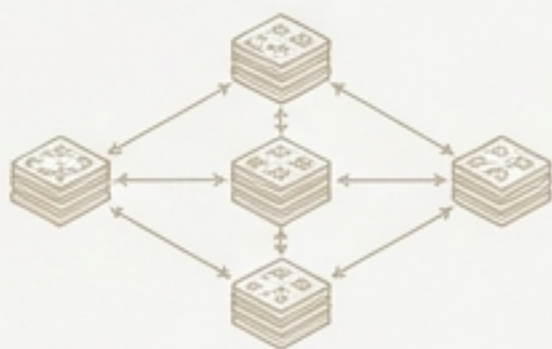
Scales to zero.
No idle compute cost.



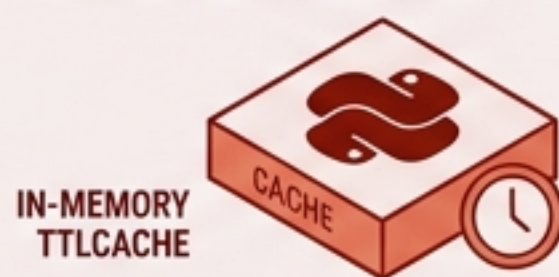
SESSION STATE

SECTION 2: SESSION STATE SCHEMATIC

Redis



Python cachetools.TTLCache



In-process memory.

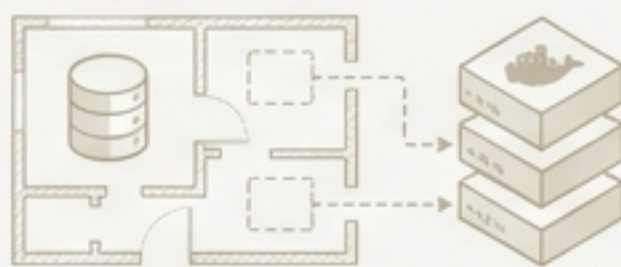
Zero external dependencies.
Zero network hop.



ARTIFACT STORAGE

SECTION 3: ARTIFACT STORAGE PLAN

PostgreSQL + Docker volumes

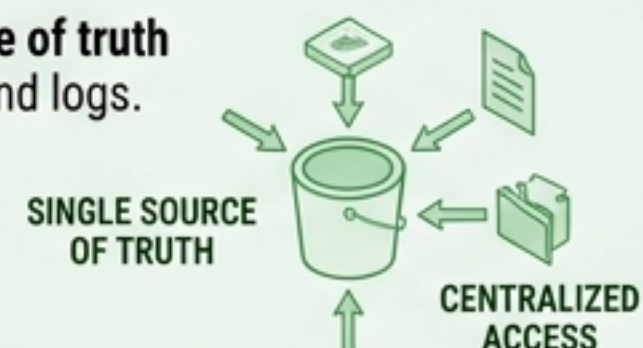


Google Cloud Storage



Single source of truth

for models and logs.



LLM ENGINE

SECTION 4: LLM ENGINE ELEVATION

Anthropic Claude

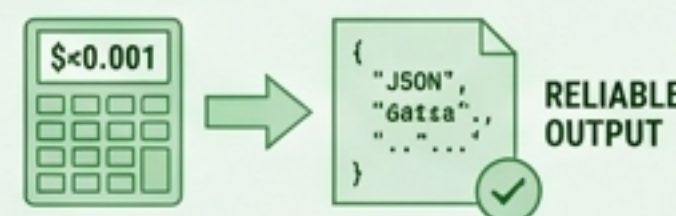


Gemini 2.0 Flash / 2.5 Flash-Lite



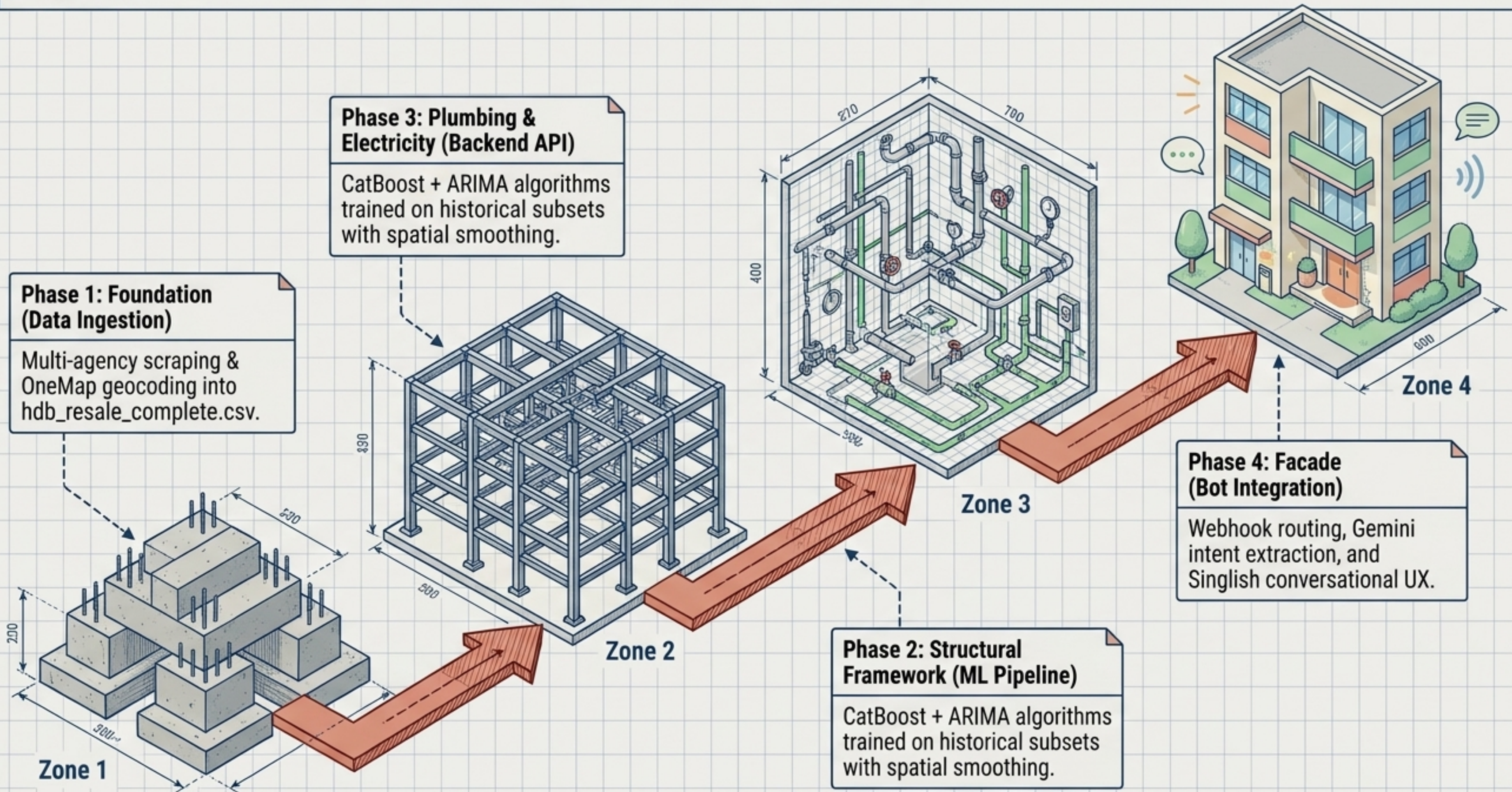
Sub-\$0.001 per-session cost.

Exceptional JSON output reliability.





ESTATE MASTER ROADMAP: SYSTEM EVOLUTION & INTERCONNECTION



THE DATA ENRICHMENT FUNNEL

LTA:
Bus stops & MRT rail exits (GeoJSON)

MOE:
Primary & Secondary school directories

HDB:
Raw resale flat prices (2017+) & property inventory

NEA:
Hawker centre polygons

OpenStreetMap:
Shopping mall coordinates (Overpass API)

The Mixer

Public OneMap Search API caching
-> Resolves exact address coordinates.

Output

hdb_resale_complete.csv
(The definitive, enriched wide-table dataset)

ML FEATURE TAXONOMY: ARCHITECTURAL FLOOR PLAN

ZONE 1

Floor area (sqm), mid-storey band, total dwelling units, remaining lease percentage, high-floor indicator.

ZONE 1 UNIT & BUILDING (THE INTERIOR)

ZONE 2

Log distances to nearest MRT, mall, hawker, and bus. Accessibility score ($0.4 \times \text{MRT} + 0.2 \times \text{Mall} + 0.2 \times \text{Hawker}$).

LOCATION & DISTANCES (THE EXTERIOR)

ZONE 3

Official quarterly HDB Resale Price Index (RPI) and its interactions with transaction year and floor area.

RPI Trend

ZONE 4

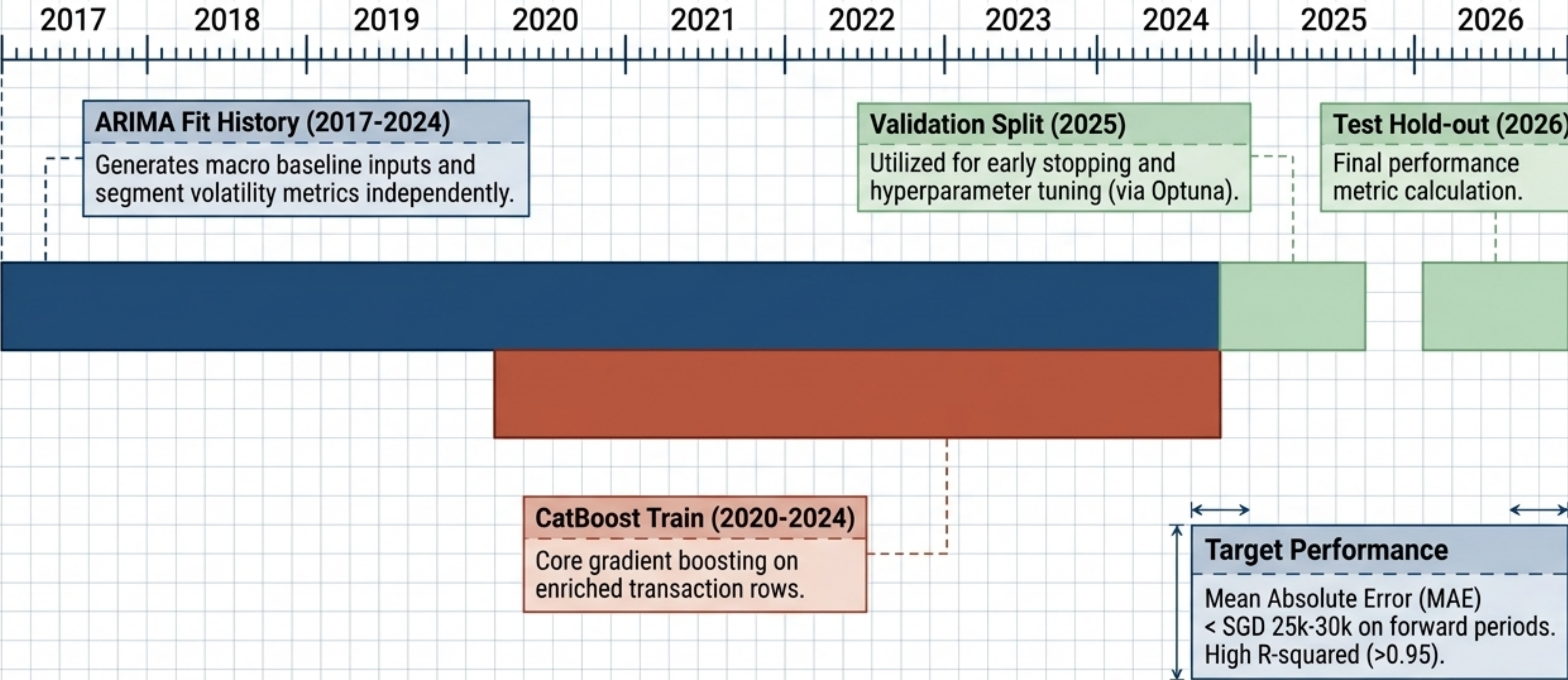
CROSS INTERACTIONS (THE LOAD-BEARING WALLS)

Area x Storey
Area x Lease Remaining
Year Completed x Floor Area

TIME-BASED SPLIT TIMELINE: ARCHITECTURAL GANTT CHART



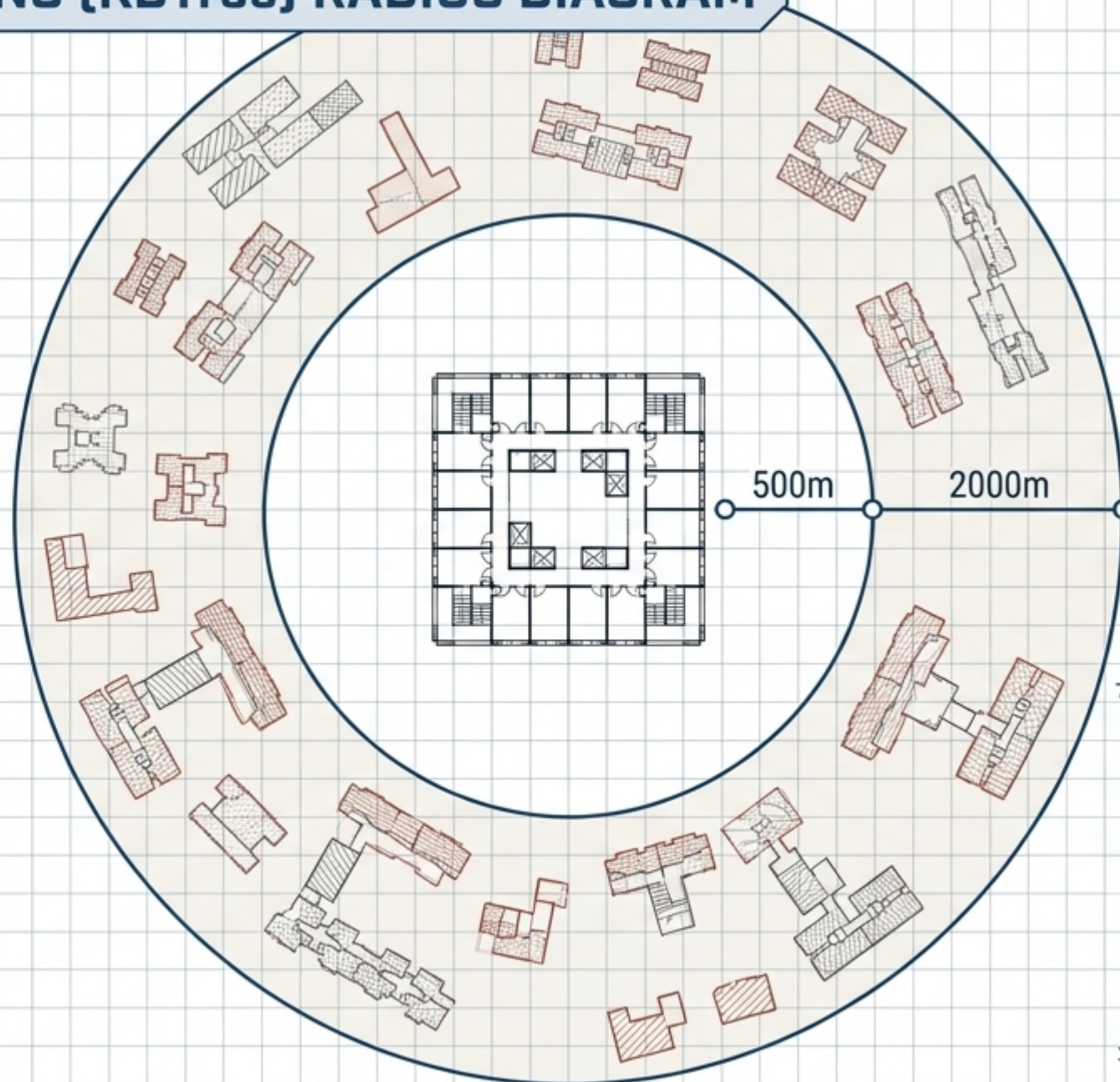
BANNED:
Random Splits



SPATIAL SMOOTHING (KDTree) RADIUS DIAGRAM

KDTree Neighborhood Encodings

- **500m & 2000m Radii:** Captures surrounding target-style signals.
- **Metrics Smoothed:** Total price encoding (spatial_500m_te) and Price-per-sqm encoding (spatial_500m_psm).



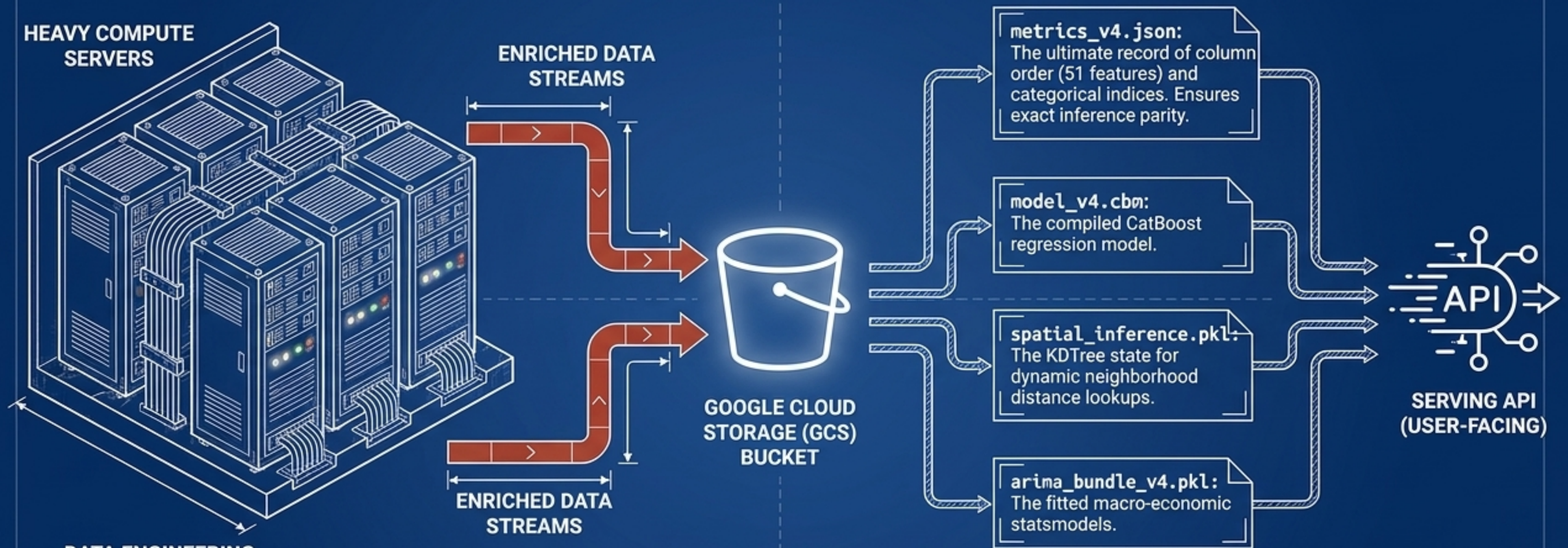
ARIMA Market Overlay

- **Segment vs Global:** Segment log-level minus global log-level for the transaction month (e.g., specific town + flat type relative to overall market).

ARIMA Market Overlay

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- **Volatility:** Historical log-price volatility (arima_seg_series_std).

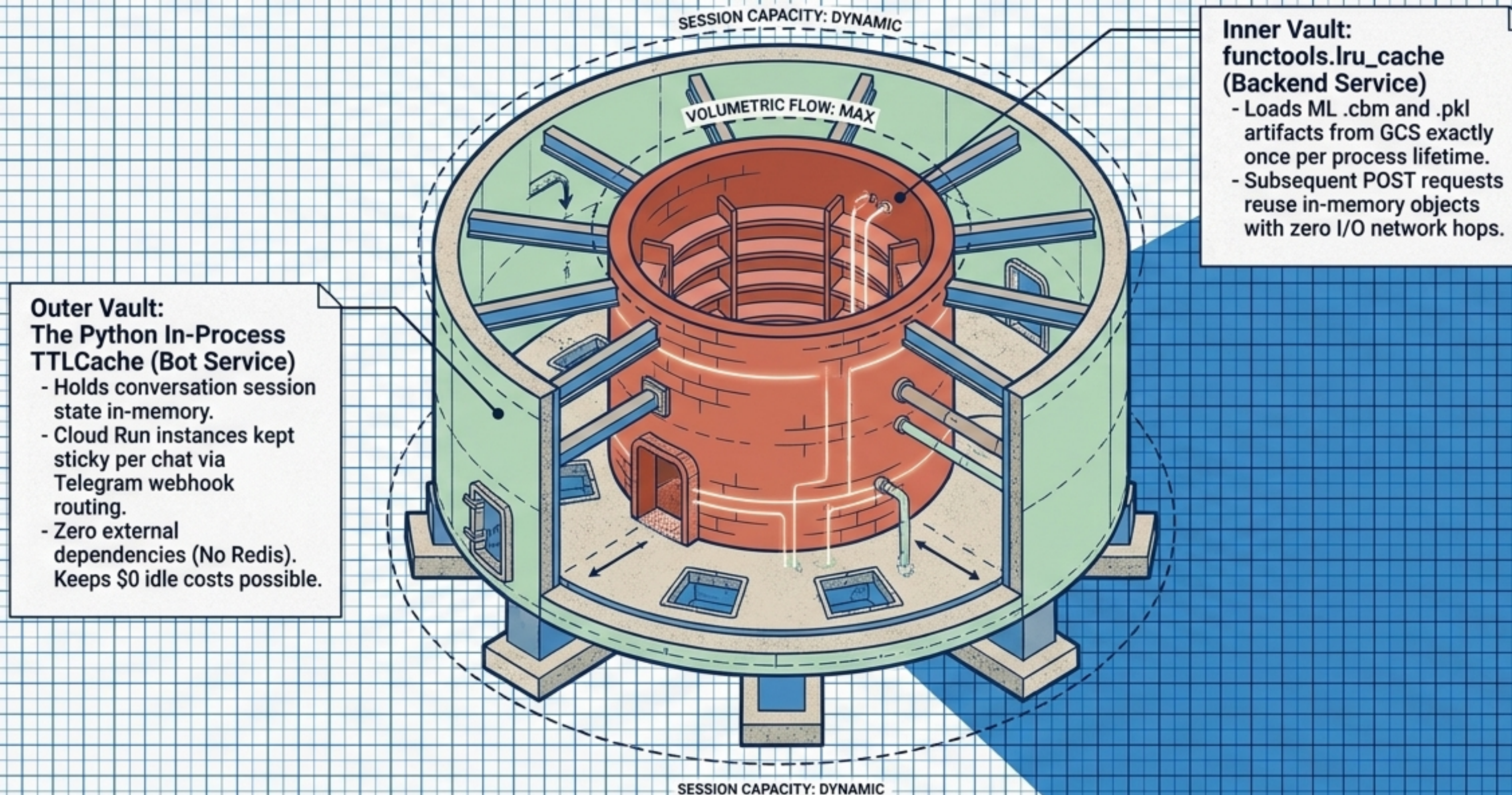
THE SINGLE SOURCE OF TRUTH



SYNTHESIS INSIGHT:

Days of data engineering and ML training are distilled into lightweight files residing in GCS, ready to be summoned into memory to serve a user in under a second.

THE INFERENCE CACHING MODEL



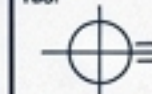
PROJECT:
HDB INFERENCE
ARCHITECTURE

REVISION NOTE

SCALE: 98/01:
2 209

SCALE:
R.T.S.

TBS:



DRAWING NO:

A-301

DATA EXPANSION ASSEMBLY LINE

STAGE 1: INPUT (POST /predict)

Only 4 required fields:
Town, Block, Storey Range,
Floor Area (sqm).

STAGE 2: EXPANSION (build_inference_pool)

Missing parameters (Flat Type, Flat
Model, Remaining Lease) are instantly
resolved via block_lookup.parquet.

block_lookup.parquet

STAGE 3: ENGINEERING

Distances, amenities, and
RPI interactions are
computed on-the-fly.

AMENITIES

RPI

RPI

DISTANCES

STAGE 4: INFERENCE

Data enters the CatBoost Pool.
The target is calculated as
 $\log(p(\text{resale_price}))$, and np.expm1
is applied to output the final
raw SGD value.

CatBoost
Pool

Target is calculated as
 $\log(p(\text{resale_price}))$.

np.expm1
applied

FINAL RAW SGD
VALUE OUTPUT

\$20000 SGD

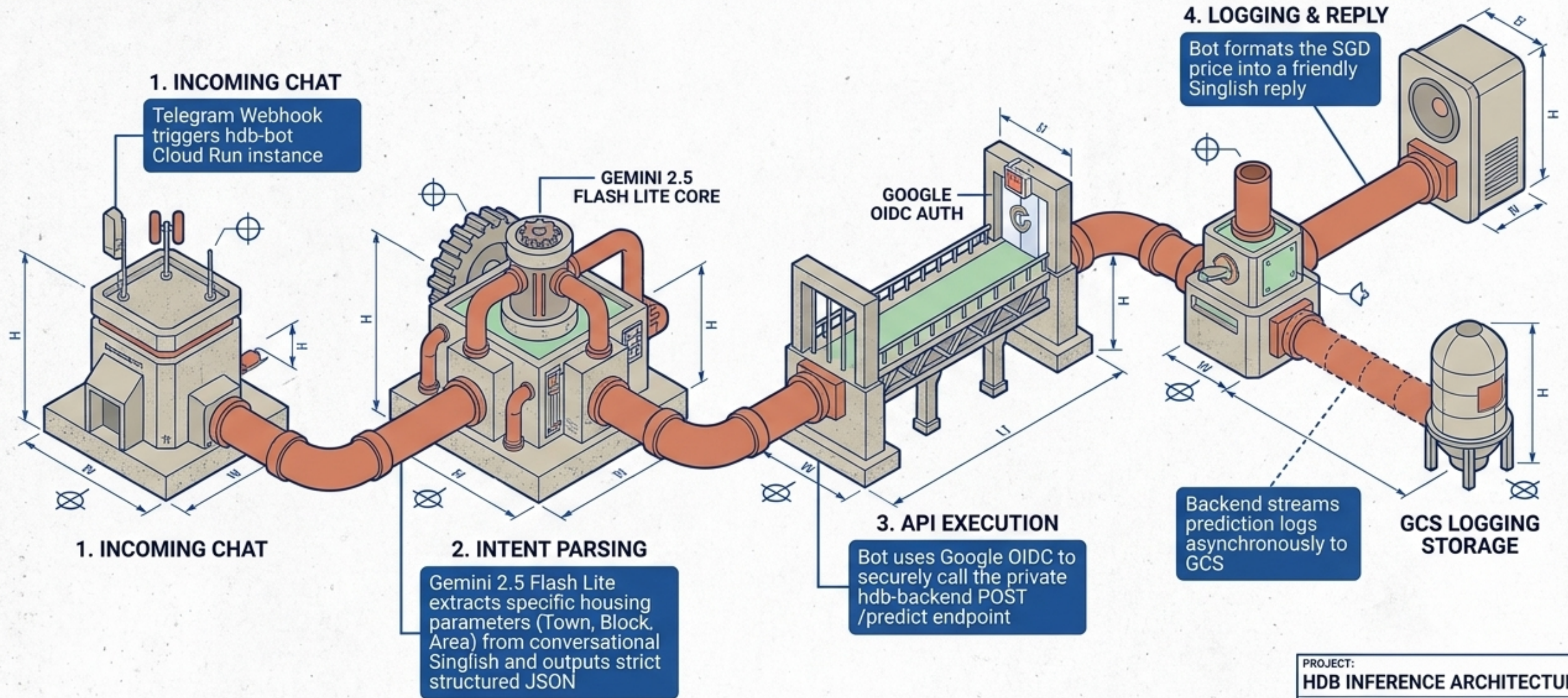
PROJECT: HDB INFERENCE ARCHITECTURE

DRAWING NO: A-302

SCALE: N.T.S.

DATE: 2023-09-22

SERVERLESS EXECUTION FLOW



PROJECT:
HDB INFERENCE ARCHITECTURE

DRAWING NO:
A-303

SCALE: **N.T.S.** DATE: **2023-09-22**

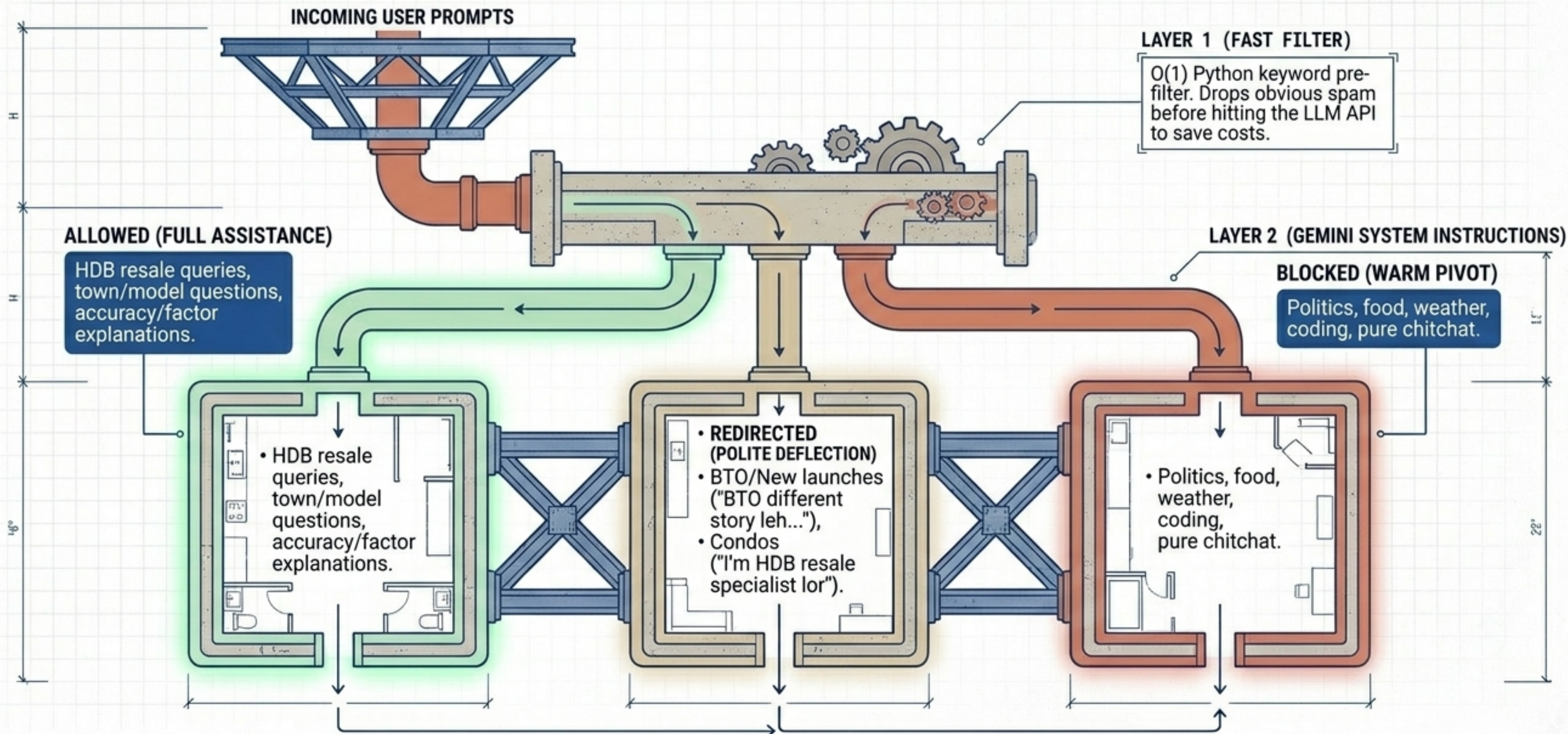
THE GUARDRAIL MATRIX

PROJECT: HDB INFERENCE ARCHITECTURE

DRAWING NO: A-304

SCALE: N.T.S.

DATE: 2023-09-22



THE SINGLISH MATRIX

CORE TONE RULES

Short paragraphs
(2-3 sentences).

Warm, never
sarcastic.

Never repeat the
same particle in
consecutive
sentences.

Always
acknowledge first.

Emoji discipline
(1 max on follow-ups).

VOCABULARY GUARDRAILS

Ah / Leh / Lor: Softeners
and context-seekers
("Which town ah?",
"Expensive leh!").

Wah / Sia / Walao:
Scaled expressions of
surprise based on price
extremes.

**Steady / Confirm plus
chop:** Affirmative
confirmations of data
capture.

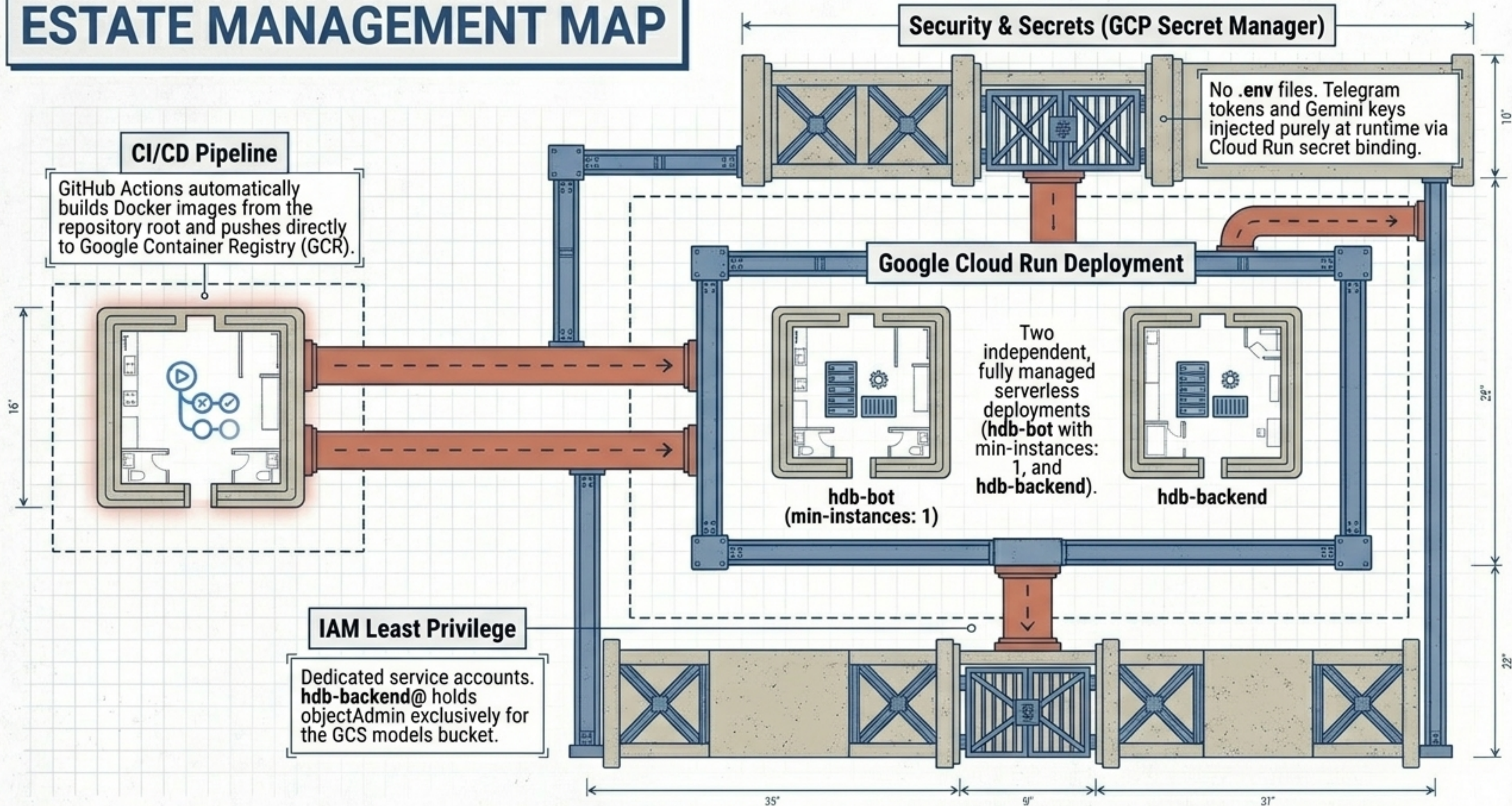
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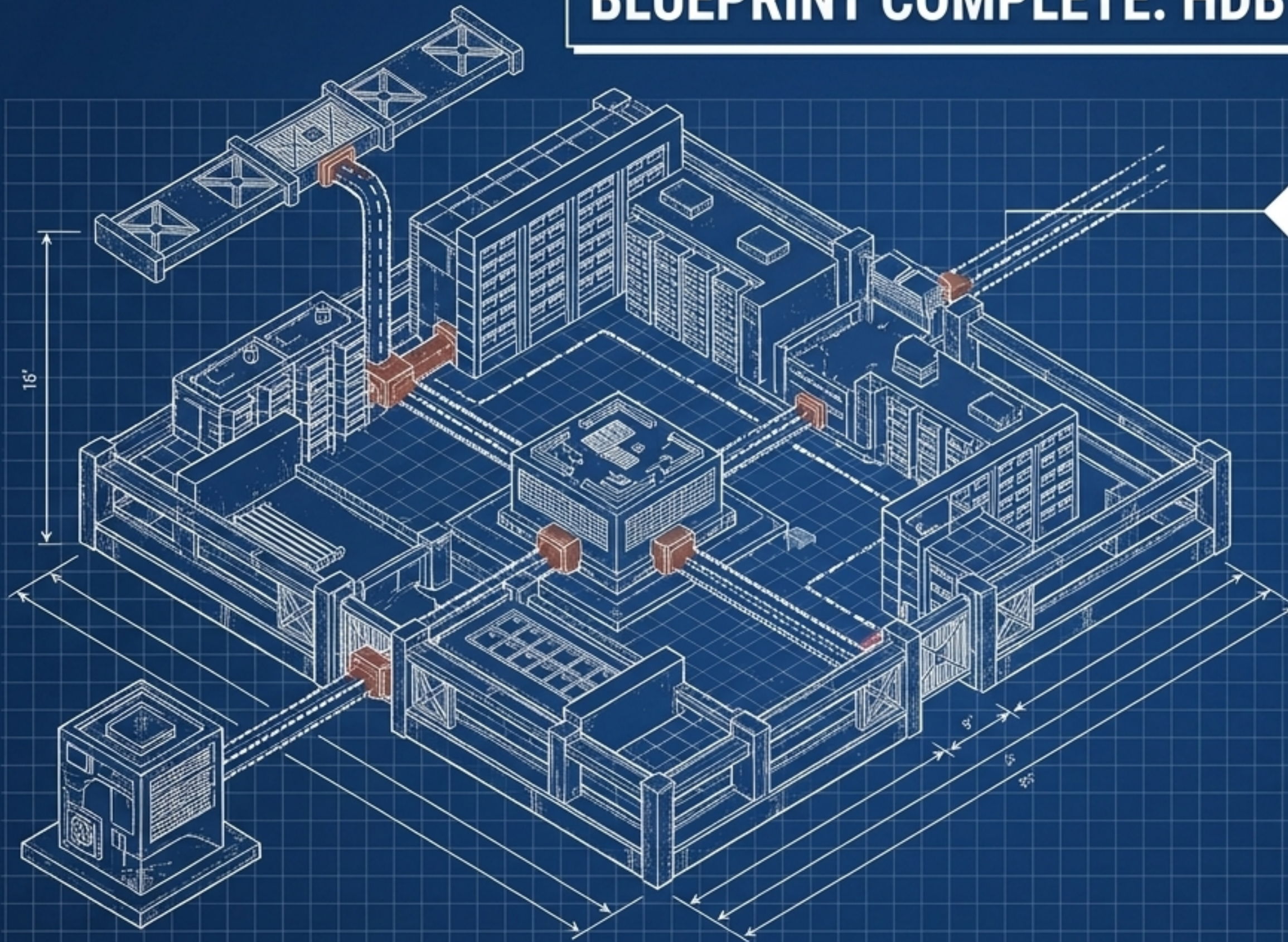
**Steady / Confirm plus
chop:** Affirmative
confirmations of data
capture.

Aiyo / Blur: Mild
facepalms for incorrect
user inputs or off-topic
redirects.

ESTATE MANAGEMENT MAP



BLUEPRINT COMPLETE: HDB BOT PIPELINE ESTATE



Monthly Operations Profile (MVP Scale)

- **Cloud Run (hdb-bot):** ~\$8.00 (1 instance, 0.5 vCPU, Always-on for TTLCache).
- **Cloud Run (hdb-backend):** ~\$15.00 (1 instance, 1 vCPU).
- **Google Cloud Storage & Secrets:** ~\$0.20 (~200MB artifacts).
- **Gemini 2.5 Flash-Lite:** Micro-cents per session token.

The Bottom Line: A state-of-the-art, machine-learning-driven natural language real estate application running continuously in production for ~\$25-35 / month.

Status: Serverless. Stateless. Scalable.
Blueprint Complete.